

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$24x + y = 48$

$6x + y = 72$

The solution to the given system of equations is (x, y) . What is the value of y ?

Correct Answer: 80

Rationale

The correct answer is **80**. Subtracting the second equation in the given system from the first equation yields $(24x + y) - (6x + y) = 48 - 72$, which is equivalent to $24x - 6x + y - y = -24$, or $18x = -24$. Dividing each side of this equation by **3** yields $6x = -8$. Substituting -8 for **6x** in the second equation yields $-8 + y = 72$. Adding **8** to both sides of this equation yields $y = 80$.

Alternate approach: Multiplying each side of the second equation in the given system by **4** yields $24x + 4y = 288$. Subtracting the first equation in the given system from this equation yields $(24x + 4y) - (24x + y) = 288 - 48$, which is equivalent to $24x - 24x + 4y - y = 240$, or $3y = 240$. Dividing each side of this equation by **3** yields $y = 80$.